

Problem 14.2.9

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \exp(-z^2)$$

Solution

f is entire since f' is defined everywhere, therefore by Cauchy's integral theorem

$$\oint_C f(z) \, dz = 0$$

Problem 14.2.11

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{2z - 1}$$

Solution

f' is not defined at $z = \frac{1}{2}$ and therefore Cauchy's integral theorem does not apply. By 14.2.(3), $\oint_C (z - z_0)^{-1} = 2\pi i$. Therefore,

$$\begin{aligned} \oint_C f(z) \, dz &= \frac{1}{2} \oint_C \left(z - \frac{1}{2}\right)^{-1} \, dz \\ &= \boxed{\pi i} \end{aligned}$$

Problem 14.2.13

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{z^4 - 1.1}$$

Solution

f' is not defined at $z = \sqrt[4]{1.1} \approx 1.02$, however this is outside the unit circle and therefore Cauchy's integral theorem applies.

$$\oint_C f(z) \, dz = 0$$

Problem 14.2.18

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{4z - 3}$$

Solution

f' is not defined at $z = \frac{3}{4}$ and therefore Cauchy's integral theorem does not apply. By 14.2.(3), $\oint_C (z - z_0)^{-1} = 2\pi i$. Therefore,

$$\begin{aligned} \oint_C f(z) dz &= \frac{1}{4} \oint_C \left(z - \frac{3}{4}\right)^{-1} dz \\ &= \boxed{\frac{\pi i}{2}} \end{aligned}$$

Problem 14.2.22

Evaluate the integral. Does Cauchy's theorem apply? Show details.

$$\oint_C \operatorname{Re} z \, dz$$

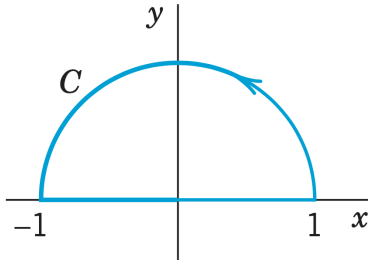


Figure 1: C

Solution

$$f(z) = \operatorname{Re} z = u(x, y) + iv(x, y)$$

$$\begin{aligned} u &= x & v &= 0 \\ u_x &= 1 & v_x &= 0 \\ u_y &= 0 & v_y &= 0 \end{aligned}$$

$u_x \neq v_y$ therefore f is not analytic and Cauchy's theorem does not apply.

$$C_1 : z_1 = e^{it}, dz_1 = ie^{it} dt, \operatorname{Re} z_1 = \cos t, \quad t \in [0, \pi]$$

$$C_2 : z_2 = t, dz_2 = dt, \operatorname{Re} z_2 = t, \quad t \in [-1, 1]$$

$$\begin{aligned}
\oint_C f(z) dz &= \oint_C f(z(t)) \dot{z}(t) dt \\
&= \oint_{C_1} f(z_1(t)) \dot{z}_1(t) dt + \oint_{C_2} f(z_2(t)) \dot{z}_2(t) dt \\
&= \int_0^\pi \cos t i e^{it} dt + \int_{-1}^1 t dt \\
&= \int_0^\pi \cos t i (\cos t + i \sin t) dt + \int_{-1}^1 t dt \\
&= i \int_0^\pi \cos^2 t dt - \int_0^\pi \cos t \sin t dt + \int_{-1}^1 t dt \\
&= \frac{i}{2} \int_0^\pi 1 + \cos 2t dt - \frac{1}{2} \int_0^\pi \sin 2t dt + \int_{-1}^1 t dt \\
&= \frac{i}{2} \left[t + \frac{\sin 2t}{2} \right]_0^\pi + \frac{1}{4} [\cos 2t]_0^\pi + \frac{1}{2} [t^2]_{-1}^1 \\
&= \frac{i}{2} [\pi] + \frac{1}{4} [0] + \frac{1}{2} [0] \\
&= \boxed{\frac{\pi i}{2}}
\end{aligned}$$

Problem 14.2.24

Evaluate the integral. Does Cauchy's theorem apply? Show details. (*Use partial fraction*)

$$\oint_C \frac{dz}{z^2 - 1}$$

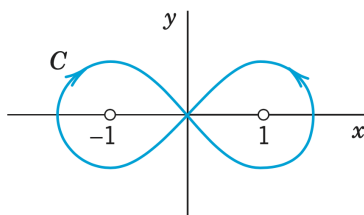


Figure 2: C

Solution

$$\frac{1}{z^2 - 1} = \frac{A}{z - 1} + \frac{B}{z + 1} \implies 1 = A(z + 1) + B(z - 1)$$

$$\begin{aligned}
z = -1: & \quad B = -\frac{1}{2} \\
z = 1: & \quad A = \frac{1}{2}
\end{aligned}$$

$$\frac{1}{z^2 - 1} = \frac{1/2}{z - 1} - \frac{1/2}{z + 1}$$

The integral becomes,

$$\oint_C \frac{dz}{z^2 - 1} = \frac{1}{2} \oint_C \frac{1}{z - 1} - \frac{1}{z + 1} dz$$

Splitting C into C_1 , the left half that circles $z = -1$ clockwise, and C_2 , the right half that circles $z = 1$ counterclockwise, we can evaluate each integral separately.

The first integral has a singularity at $z = 1$, therefore over C_1 Cauchy's theorem holds and the integral is zero. Over C_2 , Cauchy's theorem does not hold, but 14.2.(3) gives a solution of πi .

$$\frac{1}{2} \oint_C \frac{1}{z - 1} = \frac{1}{2} \left[\oint_{C_1} \frac{1}{z - 1} dz + \oint_{C_2} \frac{1}{z - 1} dz \right] = \frac{1}{2} [0 + 2\pi i] = \pi i$$

The second integral has a singularity at $z = -1$, therefore over C_1 Cauchy's theorem does not hold, but 14.2.(3) gives a solution of $-\pi i$ (due to the clockwise path). Over C_2 , Cauchy's theorem holds and the integral is zero.

$$\frac{1}{2} \oint_C \frac{1}{z + 1} = \frac{1}{2} \left[\oint_{C_1} \frac{1}{z + 1} dz + \oint_{C_2} \frac{1}{z + 1} dz \right] = \frac{1}{2} [0 - 2\pi i] = -\pi i$$

The final integral is

$$\boxed{\oint_C \frac{dz}{z^2 - 1} = 2\pi i}$$

Problem 14.3.3

Integrate by Cauchy's formula counterclockwise around the circle.

$$f(z) \frac{z^2}{z^2 - 1}, \quad |z + i| = 1.4$$

Solution

$f(z)$ has singularities at

$$z^2 - 1 = 0 \implies z_{1,2} = \pm 1$$

$$|z_{1,2} + i| = \sqrt{2} \approx 1.41 > 1.4$$

Therefore the singularities are outside the domain and Cauchy's theorem holds.

$$\boxed{\oint_C f(z) dz = 0}$$

Problem 14.3.11

Integrate counterclockwise.

$$\oint_C \frac{dz}{z^2 + 4}, \quad C : 4x^2 + (y - 2)^2 = 4$$

Solution

$f(z)$ has singularities at

$$z^2 + 4 = 0 \implies z_{1,2} = \pm 2i$$

$z_1 = 2i$ is in the center of the region, however $z_2 = -2i$ lies outside. Therefore we can split the function in to the analytic and non-analytic factors and integrate via Cauchy's integral formula.

$$\begin{aligned} \oint_C \frac{dz}{z^2 + 4} &= \oint_C \frac{1}{(z - 2i)(z + 2i)} dz \\ &= \oint_C \frac{1/(z + 2i)}{(z - 2i)} dz \\ &= \frac{1}{(z + 2i)} \Big|_{z_0=2i} \cdot 2\pi i \\ &= \frac{1}{4i} \cdot 2\pi i \\ &= \boxed{\frac{\pi}{2}} \end{aligned}$$

Problem 14.3.13

Integrate counterclockwise.

$$\oint_C \frac{z+2}{z-2} dz, \quad C : |z-1| = 2$$

Solution

$f(z)$ has a singularity at $z_0 = 2$ which lies within the region. Integrate via Cauchy's integral formula.

$$\begin{aligned} \oint_C \frac{z+2}{z-2} dz &= (z+2) \Big|_{z_0=2} \cdot 2\pi i \\ &= 4 \cdot 2\pi i \\ &= \boxed{8\pi i} \end{aligned}$$

Problem 14.3.15

Integrate counterclockwise. (compute it for two squares: one with vertices $\pm 2, \pm 2i$; the other one with vertices $\pm 4, \pm 4i$)

$$\oint_C \frac{\cosh(z^2 - \pi i)}{z - \pi i} dz, \quad C \text{ the boundary of the square with vertices } \pm 2, \pm 2, \pm 4i$$

Solution

$f(z)$ has a singularity at $z_0 = \pi i$ which lies within the region. Integrate via Cauchy's integral formula.

For the small square with vertices $\pm 2, \pm 2i$ the singularity is outside the region and Cauchy's theorem applies.

$$\oint_C \frac{\cosh(z^2 - \pi i)}{z - \pi i} dz = 0$$

For the large square with vertices $\pm 4, \pm 4i$ the singularity is inside the region and Cauchy's theorem does not apply.

$$\begin{aligned} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz &= f^{(n)}(z_0) \frac{2\pi i}{n!} \\ \oint_C \frac{\cosh(z^2 - \pi i)}{z - \pi i} dz &= \cosh(z^2 - \pi i) \Big|_{z_0=\pi i} \cdot 2\pi i \\ &= \cosh(-\pi^2 - \pi i) \cdot 2\pi i \\ &= \boxed{-2\pi i \cos \pi^2} \end{aligned}$$

Problem 14.4.1

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{\sin z}{z^4} dz$$

Solution

$f(z)$ has a singularity at $z_0 = 0$ which lies within the region.

$$\begin{aligned} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz &= f^{(n)}(z_0) \frac{2\pi i}{n!} \\ \oint_C \frac{\sin z}{z^4} dz &= (\sin z)^{(3)} \Big|_{z_0=0} \cdot \frac{2\pi i}{3!} \\ &= -\cos z \Big|_{z_0=0} \cdot \frac{\pi i}{3} \\ &= \boxed{-\frac{\pi i}{3}} \end{aligned}$$

Problem 14.4.3

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{e^z}{z^n} dz, \quad n = 1, 2, \dots$$

Solution

$f(z)$ has a singularity at $z_0 = 0$ which lies within the region.

$$\begin{aligned}
\oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz &= f^{(n)}(z_0) \frac{2\pi i}{n!} \\
\oint_C \frac{e^z}{z^n} dz &= (e^z)^{(n-1)} \Big|_{z_0=0} \cdot \frac{2\pi i}{(n-1)!} \\
&= e^z \Big|_{z_0=0} \cdot \frac{2\pi i}{(n-1)!} \\
&= \boxed{\frac{2\pi i}{(n-1)!}}
\end{aligned}$$

Problem 14.4.6

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{dz}{(z - 2i)^2(z - i/2)^2}$$

Solution

$f(z)$ has singularities at $z_1 = \frac{i}{2}$, $z_2 = 2i$

z_1 lies inside the unit circle, however z_2 does not. Separating the analytic and non-analytic factors to use Cauchy's integral formula,

$$\begin{aligned}
\oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz &= f^{(n)}(z_0) \frac{2\pi i}{n!} \\
\oint_C \frac{(z - 2i)^{-2}}{(z - i/2)^2} dz &= ((z - 2i)^{-2})' \Big|_{z_0=\frac{i}{2}} \cdot \frac{2\pi i}{1!} \\
&= 2\pi i (-2(z - 2i)^{-3}) \Big|_{z_0=\frac{i}{2}} \\
&= \boxed{-\frac{32}{27}\pi}
\end{aligned}$$

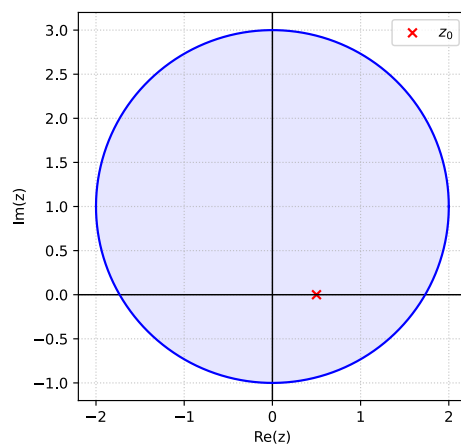
Problem 14.4.11

Integrate. Show the details. *Hint.* Begin by sketching the contour. Why?

$$\oint_C \frac{(1+z)\sin z}{(2z-1)^2} dz, \quad C : |z - i| = 2 \text{ counterclockwise}$$

Solution

$f(z)$ has a singularity at $z_0 = \frac{1}{2}$ which lies within the region

Figure 3: $C : |z - i| = 2$

$$\begin{aligned}
 \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz &= f^{(n)}(z_0) \frac{2\pi i}{n!} \\
 \oint_C \frac{(1+z)\sin z}{(2z-1)^2} dz &= \left. ((1+z)\sin z)' \right|_{z_0=\frac{1}{2}} \cdot \frac{2\pi i}{1!} \\
 &= 2\pi i (\sin z + (1+z)\cos z) \Big|_{z_0=\frac{1}{2}} \\
 &= 2\pi i \left(\sin \frac{1}{2} + \frac{3}{2} \cos \frac{1}{2} \right) \\
 &= \boxed{\pi i \sin \frac{1}{2} + 3\pi i \cos \frac{1}{2}}
 \end{aligned}$$